

Response to Request for Information: Developing a National Artificial Intelligence Action Plan

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March 15, 2025

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The Future Society welcomes the opportunity to provide input on the development of a comprehensive National Artificial Intelligence Action Plan. As a nonprofit organization with a decade of experience in governance of artificial intelligence, we believe that a strategic national approach is essential to harness AI's transformative potential while mitigating its risks.

President Trump's Executive Order on Removing Barriers to American Leadership in Artificial Intelligence has already articulated a vision to create high-paying jobs, strengthen national security, and ensure that AI development reflects American values and interests. We stand ready to support these efforts and contribute our expertise to building an AI future that puts America first. Our recommendations are therefore organized around three pillars required to cement U.S. leadership in AI: promoting its adoption, protecting its development, and planning to mitigate against new sources of technical and geopolitical risk:

- The **Promote** pillar focuses on unleashing AI's potential to foster human flourishing through accelerating American leadership in foundational AI research, strengthening domestic semiconductor manufacturing, improving government procurement processes, and advancing privacy and security science.
- The **Protect** pillar emphasizes ensuring AI systems are secure, resilient, and controllable through enhancing the U.S. AI Safety Institute, establishing robust security protocols for frontier models, strengthening protection of leading laboratories, and deploying AI for national security applications.
- The **Plan** pillar calls for new initiatives aimed at horizon-scanning and preparing for emerging risks to human flourishing, economic competitiveness, and national security – including by priming the U.S. intelligence community and industry regulators to conduct comprehensive risk assessments, incident reporting mechanisms, contingency planning, and vigilant monitoring of foreign AI projects and capabilities.

With the right balance of policies designed to advance U.S. leadership in AI, ensure system reliability and security, and prepare for contingencies, The Future Society believes that America can maintain its competitive edge in this strategic technology. The recommendations outlined in this document provide a framework for decisive action that prioritizes American innovation, security, and prosperity.

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1. PROMOTE

Unleashing AI's Potential to Foster Human Flourishing

1.1 Accelerating American Leadership in Foundational AI Research

To solidify U.S. leadership in AI and ensure equitable access to essential computational resources, it is imperative to enhance the National AI Research Resource (NAIRR). We propose a comprehensive strategy encompassing increased funding, a tiered access system, an open science mandate, and the establishment of regional centers of excellence. Our framework prioritizes American innovation by concentrating appropriate safeguards only where genuinely needed.

Increased Funding for American Competitiveness: NAIRR's computational infrastructure is essential to maintaining U.S. leadership at the frontier of AI science. Federal investment should be sustained, substantial, and prioritize support for university research centers and foundational AI science programs. Such funding would bolster the nation's capacity to perform cutting-edge research and maintain a competitive edge in the global AI landscape. Even as the United States refocuses on austerity, it will be essential to marshal state support in pursuit of strategic capabilities.

Tiered Access Supporting National Priorities: Implementing a system that provides varying levels of computational resources based on project merit, alignment with national priorities, and potential societal impact is essential. This model ensures that critical and high-potential projects receive adequate support, optimizing resource allocation and fostering innovations that address pressing societal challenges. Such an approach would democratize access to AI resources, enabling a broader spectrum of researchers and organizations to contribute to the development of AI systems, applications, and enabling technologies.

Distributed American Innovation Centers: Establishing regional centers of excellence is vital for promoting geographic diversity in AI research capabilities and talent development. These centers would ensure nationwide participation in AI advancements, mitigating regional disparities and fostering local innovation ecosystems. As the Trump administration works to unleash AI's economic potential, it should ensure these benefits are distributed throughout the American heartland – contributing to U.S. research capacity and revitalizing the U.S. economy.

1.2 Making Semiconductors in America

Sustaining American leadership in AI will require technology developers and investors to remain confident in their ability to access the world's most advanced computing power. We propose establishing a high-level task force focused on reassessing and improving the CHIPS and Science Act of 2022, developing secure supply chain certification programs, creating dedicated manufacturing capacities for defense applications, and implementing workforce development initiatives. These measures aim to address current challenges, such as project delays and financial uncertainties, and would provide for a more robust and secure semiconductor infrastructure.

Reevaluating and Enhancing the CHIPS Act: While the CHIPS and Science Act of 2022 aimed to bolster domestic semiconductor manufacturing, recent developments indicate the need for reassessment. Challenges such as project delays and financial uncertainties, as seen with Intel's manufacturing project in Ohio, have led to significant delays and financial instability. A comprehensive evaluation of the CHIPS Act's implementation is necessary to identify and address these shortcomings, ensuring that the legislation effectively supports the establishment and expansion of advanced semiconductor fabrication facilities, particularly those specializing in leading-edge GPUs.

Developing a Secure Supply Chain Certification Program: The United States should also build localized supply chains for semiconductor components used in sensitive national security domains. We propose the development of a certification program that ensures the security and reliability of semiconductor components used in critical applications. This program should build upon recent industry initiatives aimed at scaling up packaging and testing operations within the United States, thereby enhancing supply chain security and ensuring that the whole stack of next-generation chip technologies are made in America.

Establishing Dedicated Manufacturing Capacity for Defense Applications: The U.S. Department of Defense requires specialized semiconductors, including radiation-hardened and tamper-resistant designs, for its AI applications. We recommend the creation of dedicated production capacities — to include fabrication, packaging, and testing facilities — to meet these specific requirements.

Implementing Workforce Development Initiatives: A skilled workforce is essential for sustaining and advancing semiconductor manufacturing capabilities. We advocate for the

funding of specialized training programs in semiconductor design and manufacturing, with a particular emphasis on creating pathways that include security clearances. This approach will ensure that the workforce is not only technically proficient but also prepared to handle the sensitive nature of defense-related projects.

1.3 Wielding AI for Good Governance

Current procurement processes significantly hinder government adoption of AI innovations. We propose the establishment of a dedicated AI acquisition pathway, the implementation of a pre-approved vendor program, and the development of innovation sandboxes. These measures aim to streamline procurement processes, foster a dynamic marketplace, and enable controlled experimentation with AI solutions, bolstering the U.S. government's ability to deploy AI at speed and scale.

Establishing an AI-Specific Acquisition Pathway: Traditional procurement timelines are impeding the rapid adoption of innovative AI solutions. To address this, we propose the creation of a dedicated acquisition pathway tailored for AI systems. This pathway should feature expedited evaluation periods, ideally not exceeding 90 days, to ensure timely integration of AI capabilities into government operations. Such a streamlined process would enable agencies to respond swiftly to emerging challenges and technological opportunities, thereby enhancing operational efficiency and public service delivery.

Implementing a Pre-Approved Vendor Program: To facilitate quicker procurement cycles, establishing a rigorous yet accelerated qualification process for AI vendors is essential. A pre-approved vendor program would allow agencies to engage with vetted AI solution providers without the delays associated with traditional procurement procedures. This approach not only reduces administrative burdens but also encourages a more dynamic and competitive marketplace, fostering innovation and ensuring that the government has access to cutting-edge technologies.

Developing Innovation Sandboxes: Authorizing federal agencies to create innovation sandboxes—controlled environments where AI solutions can be tested on a limited scale—would significantly enhance the government's ability to experiment with new technologies. These sandboxes enable agencies to assess the efficacy and implications of AI applications without committing to full-scale procurement. This iterative approach allows for real-world testing, risk mitigation, and informed decision-making, ultimately leading to more effective and efficient AI deployments.

1.4 Pushing the Frontier of Privacy and Security Science

Advancing the frontiers of privacy and security in AI necessitates a comprehensive approach that includes dedicated research funding, continuous monitoring, and innovative encryption techniques. We propose several initiatives to scale up foundational work in privacy-preserving technologies:

Establishing a Dedicated NSF Evaluation Research Program: Investing in foundational research to develop rigorous AI evaluation methodologies is crucial. We propose the creation of a dedicated funding stream within the National Science Foundation (NSF) to support this endeavor. This program should focus on developing metrics and frameworks that assess AI systems' performance, transparency, and societal impact. Such an initiative would ensure that AI systems are evaluated comprehensively, fostering trust and reliability in their deployment.

Defining Continuous Monitoring Standards: To maintain the integrity and performance of deployed AI systems, it is essential to establish best practices for continuous monitoring, defense in depth, and security standards across whole-lifecycle development. We recommend developing standards that enable the detection of performance drift and emergent behaviors in AI applications. These standards should be adaptable across various domains, ensuring that AI systems remain aligned with their intended functions and ethical considerations throughout their lifecycle.

Launching a Homomorphic Encryption Research Initiative: Balancing data utility with privacy protection is a significant challenge in AI development. We advocate for a five-year research program focused on practical applications of fully homomorphic encryption for government and critical infrastructure. This initiative would enable data processing without exposing sensitive information, thereby enhancing security and privacy in AI operations.

Establishing Federated Learning Testbeds: To facilitate collaboration without compromising data privacy, we propose the creation of cross-agency testbeds for federated learning approaches. Federated learning allows multiple organizations to train AI models collectively without centralizing their datasets, preserving data confidentiality. Implementing such testbeds would promote innovation while respecting privacy constraints, particularly in sectors like healthcare and finance.

Initiating a Privacy-Preserving AI Challenge Program: Encouraging breakthroughs in privacy-preserving machine learning techniques requires targeted incentives. We recommend launching competitive challenges with substantial prizes to stimulate innovation in this area. Such a program would attract diverse talent and accelerate the development of methodologies that safeguard privacy without hindering the utility of AI systems.

2. PROTECT

Ensuring AI Systems are Secure, Resilient, and Controllable

2.1 Building an AI Safety Institute That Puts America First

The U.S. AI Safety Institute plays an indispensable role in safeguarding the development and deployment of AI systems. To maximize its impact, we propose the following enhancements to its mandate:

Establishing a Protected Disclosure Framework: Encouraging transparency in reporting AI safety incidents, hazards, and near-accidents is essential for continuous improvement and risk mitigation. We recommend the creation of a protected disclosure framework that allows companies to share information about safety incidents without fear of regulatory repercussions. This mechanism would foster a culture of openness and collective learning, leading to more robust AI systems.

Developing Measurable Safety Benchmarks: Standardizing safety measures across AI systems is vital for consistent evaluation and trustworthiness. We propose that the AI Safety Institute — and the National Institute of Standards and Technology (NIST) more broadly — create measurable voluntary safety benchmarks tailored to various AI capabilities and deployment contexts. This collaboration would ensure that safety standards are both rigorous and applicable, providing clear guidelines for developers and users alike. Notably, NIST has been instrumental in developing the AI Risk Management Framework, which serves as a foundation for such benchmarks.

2.2 Ensuring Frontier Model Security and Reliability

For American AI to remain the gold standard globally, consumers of U.S.-designed AI products need to feel confident in the technology's ability to function as intended and with minimal risk of hallucination or malfunction. A comprehensive approach to frontier AI safety and reliability will include updating federal guidelines, enhancing supply chain security, and adopting secure-by-design principles throughout the whole lifecycle of AI development.

Updating OMB Circular A-130 to Address AI System Security: The rapid integration of AI into federal operations necessitates a revision of the Office of Management and Budget's (OMB) Circular A-130, which establishes policy for information governance, acquisitions, records management, open data, workforce, security, and privacy. Updating this circular to include explicit requirements for AI system security, testing, and monitoring would ensure that agencies adopt consistent and comprehensive practices in managing their adoption and use of AI technologies. This update should emphasize a shift from viewing security and privacy requirements as mere compliance exercises to understanding them as crucial components of a strategic, continuous risk-based program.

Establishing Clear, Predictable Risk Guidelines: The United States should lead the development of industry-standard verifiable "risk thresholds" for AI model releases, drawing from established risk management practices in sectors like nuclear power and aviation. These would provide regulatory certainty for American companies, replacing the previous administration's ambiguous "safety" mandates with specific benchmarks that pave the way for rapid adoption in low-risk use-cases. These thresholds should define maximum acceptable risk levels (such as 0.1% probability of causing the death of 1000 people) expressed in absolute terms rather than relative comparisons.

Cementing U.S. Leadership of the Responsible AI in the Military (REAIM) Framework: The United States should continue proactively shaping the Responsible AI in the Military (REAIM) framework to reflect U.S. national interests while establishing practical international norms. Participation in forums like the REAIM summit, which has seen endorsements from more than 60 countries, allows for the development of guidelines that govern the responsible use of AI in military contexts. By doubling down on U.S. leadership of global discussions on AI risk and safety, the Trump administration can ensure that emerging international norms and standards align with American ethical standards and strategic interests.

Enhancing Software Supply Chain Security: To mitigate risks associated with AI components in critical systems, it is essential to implement rigorous verification requirements, including provenance tracking of training data and models. This approach ensures the integrity and reliability of AI systems by addressing potential vulnerabilities in the software supply chain. Such measures align with broader federal efforts to enhance cybersecurity and protect against threats to digital infrastructure.

Developing Secure-by-Design Frameworks: Promoting architectural approaches that incorporate safety and control mechanisms is essential to ensure successive generations of AI systems remain resilient and controllable. A secure-by-design philosophy ensures that security considerations are integrated from the inception of system development, rather than being retrofitted. This proactive approach aligns with the objectives outlined in recent executive actions aimed at enhancing cybersecurity and AI safety across federal agencies.

2.3 Enhancing Security at Leading Laboratories

Establishing Minimum Physical Security Standards: AI research facilities, especially those housing advanced computing clusters, must adhere to stringent physical security protocols to prevent unauthorized access and potential sabotage. The Department of Homeland Security should conduct a comprehensive assessment of existing guidelines that outline minimum security requirements for data centers servicing U.S. government clients – including access controls, surveillance systems, and personnel vetting procedures. These standards should be tailored to the sensitivity of the data and the criticality of the infrastructure involved.

Developing AI-Specific Cybersecurity Protocols: The unique risks and vulnerabilities inherent in frontier AI systems necessitate specialized cybersecurity measures. Threats such as model theft, data poisoning, and adversarial attacks require targeted defenses. Through jailbreaking, prompt engineering, and side-channel attacks, malicious actors have continued to demonstrate a sophisticated ability to manipulate inputs to deceive AI models. We recommend adopting Level 4-5 Security Standards as defined by the RAND Corporation. NIST could create detailed security guidelines equivalent to the Level 4 and 5 benchmarks described by RAND, specifically tailored for advanced AI models and their development environments.

Establishing Dedicated Threat Intelligence Sharing Channels: Timely dissemination of threat intelligence is necessary for preempting and mitigating security incidents. We propose the creation of dedicated communication channels between government intelligence agencies and leading AI research laboratories. This collaboration would facilitate sharing real-time threat information, enabling proactive defense measures. Recent initiatives, such as the playbook introduced by the Cybersecurity and Infrastructure Security Agency (CISA) for reporting AI security threats, could serve as a template for such an initiative.

Implementing Expedited Clearance Processes for Key Personnel: Access to sensitive threat information is vital for key personnel within AI research facilities. We recommend implementing expedited security clearance processes to ensure that essential staff can promptly receive and act upon classified intelligence. This approach would enhance any given facility's ability to respond to emerging threats and align with national security objectives.

2.4 Deploying AI to Protect the Homeland

Just as AI systems must be protected from foreign threats and malign actors that would seek to misuse them, AI systems also offer significant potential to enhance U.S. security posture. We recommend:

Developing AI-Powered Biodefense Monitoring Systems: The early detection of biological threats, including engineered pathogens, is one of AI's clearest national security use-cases. We recommend investing in AI-driven surveillance systems capable of analyzing vast datasets to identify anomalies indicative of potential biological hazards. Such systems can enhance our preparedness and response strategies by providing timely alerts and actionable intelligence. Moreover, it is becoming increasingly important to safeguard the bioeconomy against digital threats.

Establishing a Cyber Defender Advantage Program: To maintain a strategic edge in cybersecurity, it is essential to provide authorized defense entities with differentiated access to advanced AI tools. We propose the creation of a Cyber Defender Advantage Program, which would equip cybersecurity professionals with state-of-the-art AI capabilities, enabling proactive threat detection and mitigation. This initiative should include robust oversight mechanisms to ensure ethical use and prevent potential misuse of these powerful technologies. Recent developments, such as the introduction of AI-

powered tools that automate threat defenses, exemplify the potential benefits of integrating AI into cybersecurity operations.

Enhancing Critical Infrastructure Protection through AI: The security of industrial control systems and other critical infrastructure is vital to national resilience. Deploying AI-enhanced monitoring solutions can significantly improve our ability to detect anomalies and preempt potential attacks on these essential systems. Collaborations between technology companies and national agencies, such as Google's recent partnership with Australia's CSIRO – to bolster cybersecurity for critical infrastructure – is a primary example of how states have already begun wielding AI to safeguard vital assets.

3. PLAN

Horizon-Scanning and Preparing to Tackle Emerging Risks

3.1 Assessing Risks to U.S. National Security

Developing a Comprehensive Autonomous Agent Risk Framework: The increasing autonomy of AI systems necessitates a robust framework to assess their security implications. We recommend the development of comprehensive methodologies that evaluate risks associated with autonomous AI agents, including their potential misuse and unintended behaviors. This framework should consider ethical guidelines to prevent the creation of autonomous "killer robots," aligning with industry perspectives that emphasize the importance of ethical considerations in military AI applications.

Establishing Continuous Cybersecurity Capability Evaluations: Advanced AI systems can both enhance and threaten cybersecurity through vulnerability discovery, exploit development, and evasion tactics. We propose ongoing assessments of AI capabilities in these areas to ensure that defensive measures evolve in tandem with offensive potentials. The establishment of dedicated councils, such as the FCC's national security council, underscores the importance of maintaining technological superiority and addressing cyber threats from foreign adversaries, particularly in critical technologies like AI.

Implementing a Dual-Use Research Evaluation Mechanism: Advanced AI research often possesses dual-use potential, where technologies intended for beneficial purposes

could be repurposed for malicious activities. We advocate for the creation of formal processes to evaluate the dual-use nature of AI research prior to publication, balancing scientific advancement with security considerations. This aligns with concerns raised by AI leaders about the potential misuse of AI by malicious actors to cause significant harm, such as creating biological or chemical threats. It will likewise be prudent to evaluate the evolution of open source AI models globally, as these could have spillover effects with implications for U.S. national security.

Funding a Capability Forecast Program: Anticipating future AI capabilities and their national security implications is vital for proactive policy development. We recommend funding systematic research programs that forecast AI advancements over 3- to 5-year horizons, enabling the development of strategies to mitigate emerging risks. The establishment of organizations like the National Security Commission on Artificial Intelligence (NSCAI) highlights the importance of preparing for future technological challenges to maintain national competitiveness.

3.2 Incident Reporting and Information Sharing

Establishing a Voluntary Information Sharing and Analysis Center (ISAC): Creating a protected forum for sharing technical details of AI misuse incidents and near-misses is crucial for collective learning and risk mitigation. We recommend the establishment of a voluntary Information Sharing and Analysis Center (ISAC) dedicated to AI. This center would facilitate collaboration among industry stakeholders, enabling them to confidentially share information about vulnerabilities, threats, and incidents. Such a platform would mirror existing initiatives in other sectors, promoting a unified approach to AI security. Notably, OpenAI's Safety and Security Committee has proposed the creation of an ISAC to enhance threat intelligence sharing within the AI sector – a role currently filled by private sector-led initiatives like the Frontier Model Forum.

Implementing Anonymized Reporting Mechanisms: To encourage the reporting of concerns related to AI systems without fear of reprisal, it is essential to develop efficient channels for reporting security concerns that protect American businesses from liability while gathering essential intelligence on emerging threats. Drawing inspiration from the Federal Aviation Administration's confidential Aviation Safety Reporting System (ASRS), a similar approach can be adapted for AI incident reporting. This system would ensure that individuals can safely disclose information about AI-related risks or incidents, thereby contributing to a culture of safety and accountability.

Developing a Standardized Incident Classification Framework: Establishing a common taxonomy for categorizing and prioritizing different types of AI incidents is vital for consistent reporting and analysis. We advocate for the development of a standardized incident classification framework that encompasses various AI-related risks, including biases, security vulnerabilities, and system failures. Such a framework would facilitate clearer communication among stakeholders and support the development of targeted mitigation strategies. The OECD's report on a common reporting framework for AI incidents provides a valuable template for this initiative.

Cataloguing AI Incidents and Conducting Regular Trend Analysis: Periodic assessments of incident patterns are essential to identify emerging risks and opportunities for mitigation. By analyzing trends in reported AI incidents, stakeholders can proactively address systemic issues and enhance the resilience of AI systems. Initiatives like the AI Incident Database (AIID), which catalogs AI-related harm events; and the MIT AI Risk Repository, which taxonomizes modes of AI risk, serve as valuable resources for understanding and addressing potential risks associated with AI.

3.3 Contingency Planning and Rapid Response Mechanisms

Establishing a Rapid Response Council for AI Threats: The President's Council of Advisors on Science and Technology should establish a subcommittee of cross-disciplinary experts capable of assessing emerging AI-related threats to critical infrastructure or national security. Running from 2018-2021, the Congressionally-appointed National Security Committee on AI convened experts from government, industry, and academia to forecast potential opportunities and risks from increasingly capable AI systems. In 2025, the U.S. Executive Branch would be well-served by a similar advisory body – staffed with experts from government, industry, civil society, and the science community – which could rapidly assess emerging risks and, where appropriate, evaluate possible response options.

Developing AI Incident Response Playbooks: The U.S. Cybersecurity and Infrastructure Security Agency (CISA) should collaborate with industry stakeholders to develop comprehensive AI Incident and Vulnerability Response Playbooks that provide standardized protocols for addressing AI-related emergencies. These playbooks, modeled after existing cybersecurity response frameworks, would incorporate advanced detection techniques to analyze potential cascading effects of AI failures before they occur.

3.4 Monitoring Foreign Capabilities

Developing Specialized Technical Intelligence Teams: To effectively assess foreign AI developments, the U.S. intelligence community could consider establishing and expanding specialized teams focused on technical intelligence related to foreign AI capabilities. These teams would be responsible for analyzing advancements in AI systems by other nations – particularly the People’s Republic of China – and their potential to impact U.S. national security and economic competitiveness. While the United States has dedicated significant resources to foreign technical monitoring, establishing similar initiatives focused squarely on AI could help better anticipate and counter potential threats arising from a generational suite of technologies.

Leveraging Open-Source Intelligence (OSINT) to Monitor Foreign Threats: A systematic approach to tracking and analyzing published AI research worldwide is crucial for identifying global capability trends. Investing in open-source intelligence programs that monitor international AI research publications can provide valuable insights into emerging technologies, capabilities, and research methodologies. An Open Translation and Analysis Center (OTAC) or similar initiative would enable the United States to stay abreast of global developments and maintain a competitive edge in AI innovation, and would provide a source of unclassified intelligence products that could be shared easily with technology developers and likeminded partners.

Establishing an AI-Specific Technology Assessment Office: Creating a dedicated entity within the Office of Science and Technology Policy (OSTP) to synthesize intelligence on foreign AI capabilities would centralize efforts to evaluate and respond to global AI developments. This office would collaborate with existing bodies, such as the National Security Commission on Artificial Intelligence (NSCAI) to consolidate expertise and bolster U.S. capacity to address the challenges posed by AI.